

A Minimal Copper-Plate Storage-Scheduling Benchmark for Differential Dynamic Programming

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I. COPPER-PLATE STORAGE-SCHEDULING MODEL

A. Problem Statement

We consider the smallest multi-period scheduling problem that still exhibits the two features a differential dynamic programming (DDP) solver must contend with in an MPOPF setting: an inter-temporal state coupled by a discrete-time dynamic, and a nonlinear equality constraint that must hold in every period. The network is collapsed to a single node—a copper plate—so that the model retains the temporal structure of the MPOPF of [1] while carrying none of its spatial structure.

A single storage device and a single grid connection serve a known demand p_L^t over a horizon of T periods of length Δt . Following the sign convention of the branch-flow formulation, $P_B^t > 0$ denotes discharging and $P_B^t < 0$ charging. All quantities are in per unit on a common power base.

Remark on indexing. Unlike the end-of-period convention used for B_j^t in the tADMM formulation, B^t here denotes the energy at the *start* of period t , so that B^1 is given and B^{T+1} is the terminal energy. This is required by the optimal-control form of Section I-C, in which the state entering a stage must be the quantity the stage acts upon.

B. Optimization Problem

The complete problem is stated in one place below; the paragraphs that follow comment on its parts but do not restate them. The decision variables are the grid import P_{Subs}^t , the storage power P_B^t and the stored energy B^t , over $t \in \mathcal{T} = \{1, \dots, T\}$, with B^{T+1} the terminal energy:

$$\min_{P_{\text{Subs}}, P_B, B} \sum_{t=1}^T \left[c^t P_{\text{Subs}}^t + C_B (P_B^t)^2 \right] \Delta t + w (B^{T+1} - B^1)^2 \quad (1a)$$

$$\text{s.t. } P_{\text{Subs}}^t + P_B^t - p_L^t = 0, \quad t \in \mathcal{T} \quad (1b)$$

$$B^1 = B_0 \quad (1c)$$

$$B^{t+1} = B^t - P_B^t \Delta t, \quad t \in \mathcal{T} \quad (1d)$$

$$P_{\text{Subs}}^t \geq 0, \quad t \in \mathcal{T} \quad (1e)$$

$$P_B^{\min} \leq P_B^t \leq P_B^{\max}, \quad t \in \mathcal{T} \quad (1f)$$

$$B^{\min} \leq B^{t+1} \leq B^{\max}, \quad t \in \mathcal{T} \quad (1g)$$

Objective (1a). This is the objective of the tADMM formulation [1] unchanged: the cost of energy drawn from the substation at the time-varying price c^t , plus a small quadratic charge on battery throughput with coefficient C_B , whose role there is to discourage unnecessary cycling. Only the terminal term is new. Note that no quadratic term on P_{Subs}^t is needed: $C_B > 0$ alone makes the objective strictly convex on the feasible set, which is what renders the instance solvable in

closed form and therefore usable as a reference. Concretely, with $n_u = 2$ and $n_c = 1$ the balance Jacobian is $[1 \ 1]$, its null space is spanned by $(1, -1)$, and the reduced Hessian along that direction evaluates to C_B .

Power balance (1b). With the network collapsed to a single node, the nodal balance of [1] reduces to a single equality per period.

Storage (1c)–(1d). A lossless energy reservoir, coupled sequentially in time. This is the only coupling between periods. No round-trip efficiency term appears, and its absence is deliberate: a single η multiplying a signed P_B^t cannot be correct in both directions, since charging stores $\eta_c |P_B^t|$ whereas discharging P_B^t into the bus must drain P_B^t / η_d from the device, and η and $1/\eta$ agree only at $\eta = 1$. Representing losses honestly requires split variables $P_c^t, P_d^t \geq 0$ with $B^{t+1} = B^t + \eta_c P_c^t \Delta t - P_d^t \Delta t / \eta_d$, which is the route taken if losses are wanted later (see the remark closing Section I-C). The tADMM formulation of [1] models a lossless battery, so this benchmark does the same and drops the symbol rather than carrying it at unity.

Operational limits (1e)–(1g). Introduced one at a time. Substation import carries no upper limit; as in [1] it is only held non-negative, so that no energy is exported upstream. The *base instance* is the equality-constrained problem (1a)–(1d) alone, with all three bounds dropped; it is studied on its own before any inequality is added. Fig. 2 shows the instance solved with all three imposed, at $T = 6$; recall that $P_B^t > 0$ discharges, so a negative bar there is the battery charging.

On the terminal energy. The natural requirement that the device return to its initial state of charge, $B^{T+1} = B^1$, is imposed here as the quadratic penalty in (1a) rather than as a hard equality. The reason is a restriction of the solver rather than of the model: the DDP formulation of Section I-C admits a single stagewise constraint function shared by all stages, so a terminal-only equality cannot be expressed. The penalty is nevertheless *exact* in the variables it uses, since $B^{T+1} = B^T - P_B^T \Delta t$ is available from the stage- T state and control; the weight w therefore trades terminal energy against operating cost without introducing any approximation of the dynamics.

C. Optimal-Control Form

Problem (1a)–(1d) is restated as a discrete-time optimal control problem over $N = T$ stages, in the form required by DDP:

$$\begin{aligned} \min_{x,u} \quad & \sum_{t=1}^{N-1} \ell(x^t, u^t) + \ell^F(x^N, u^N) \\ \text{s.t.} \quad & x^1 = \hat{x}^1, \\ & x^{t+1} = f(x^t, u^t), \quad t \in \{1, \dots, N-1\}, \\ & c(x^t, u^t) = 0, \quad t \in \{1, \dots, N\}. \end{aligned} \quad (2)$$

States and controls. The state carries the stored energy and an explicit time index τ ; the control carries the two power set-points:

$$x^t = (B^t, \tau^t)^\top \in \mathbb{R}^2, \quad u^t = (P_{\text{Subs}}^t, P_B^t)^\top \in \mathbb{R}^2, \quad (3)$$

with $\hat{x}^1 = (B_0, 1)^\top$.

The time index is a state out of necessity, not modeling preference. The DDP formulation (2) admits one stage cost ℓ and one constraint function c for the entire horizon, so per-period data cannot be indexed and must instead be recovered from the state. We therefore carry τ and represent the period-varying data by the Lagrange interpolants through the nodes $\tau = 1, \dots, T$,

$$p_L(\tau) = \sum_{k=1}^T p_L^k \prod_{\substack{m=1 \\ m \neq k}}^T \frac{\tau - m}{k - m}, \quad (4)$$

and likewise for the price $c(\tau)$. These are exact at every node and infinitely differentiable, as DDP requires; their degree grows with T , which bounds the horizons for which the device is practical. Fig. 1 plots them against the data they interpolate.

Dynamics, costs and constraint.

$$f(x, u) = (x_1 - u_2 \Delta t, x_2 + 1)^\top, \quad (5)$$

$$c(x, u) = u_1 + u_2 - p_L(x_2), \quad (6)$$

$$\ell(x, u) = [c(x_2) u_1 + C_B u_2^2] \Delta t, \quad (7)$$

$$\begin{aligned} \ell^F(x, u) = & [c^T u_1 + C_B u_2^2] \Delta t \\ & + w(x_1 - u_2 \Delta t - B_0)^2. \end{aligned} \quad (8)$$

The terminal cost (8) depends on u^N as well as x^N , which is what allows the terminal energy B^{T+1} to appear exactly. The bounds (1e)–(1f) map directly onto control limits $b^L \leq u^t \leq b^U$. The energy bound (1g) is a state bound, which the formulation does not admit; it is introduced by appending a slack s^t to the control vector and replacing (1g) with

$$(x_1 - u_2 \Delta t) - B^{\min} - s^t = 0, \quad 0 \leq s^t \leq B^{\max} - B^{\min}, \quad (9)$$

appended as a second row of c , so that $n_c = 2$ and $n_u = 3$.

Remark (adding conversion losses). The split-variable storage model of Section I maps into this form without difficulty: $P_c^t, P_d^t \geq 0$ become two controls carrying native lower bounds, the balance (1b) becomes $P_{\text{Subs}}^t + P_d^t - P_c^t - p_L^t = 0$, and the dynamics stay linear, so $n_u = 3$ against $n_c = 1$ and the rank condition is comfortably met. The difficulty is elsewhere. Non-simultaneity, $P_c^t P_d^t = 0$, is normally left implicit on the

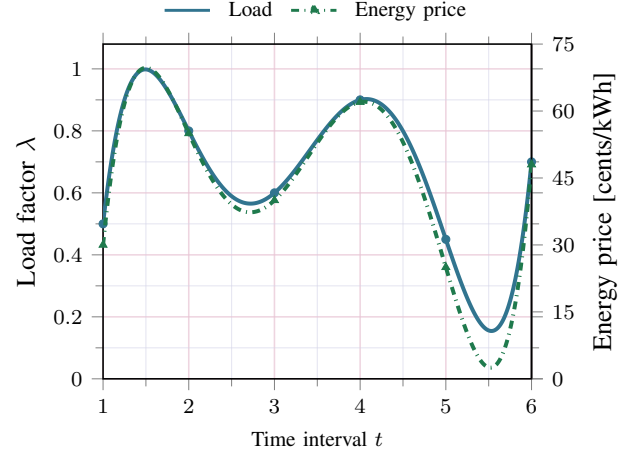


Fig. 1. Input profiles over the $T = 6$ horizon: load factor on the left axis, energy price on the right, each specified at the T horizon points. Markers are the data; the curves are the Lagrange interpolants (4) through them, and it is the curves, not the markers, that the solver evaluates. The two agree exactly at every node, by construction, but between nodes they need not be well behaved and already are not — the load interpolant reaches 0.998 against a node maximum of 0.9, and the price interpolant 69.7 cents/kWh against 62. Nothing samples those excursions here, because the time-index state advances by exactly one per stage and so only ever lands on a node; but the interpolant's derivatives at the nodes are what the backward pass consumes, and its degree is $T - 1$. This is the device that does not survive a realistic horizon.

TABLE I
INSTANCE DATA

Interval t	1	2	3	4	5	6
Demand p_L^t	1.0	1.6	1.2	1.8	0.9	1.4
Energy price c^t	0.30	0.55	0.40	0.62	0.25	0.48
Period length Δt	1	Initial energy B_0				2.0
Throughput C_B	0.5	Terminal weight w				5.0
<i>Bounds, constrained variants only</i>			$T = 3$		$T = 6$	
$P_B^{\min}, P_B^{\max}$			−0.5, 0.10		−0.5, 0.30	
$B^{\min}, B^{\max}$			1.98, 2.05		1.85, 2.10	

All quantities are in per unit on a common power base. With $\Delta t = 1$, power and energy are numerically interchangeable. The $T = 3$ instance uses the first three intervals of the series above; $T = 6$ uses all six. Substation import is unlimited above in every variant, subject only to $P_{\text{Subs}}^t \geq 0$. The bounds apply only to the constrained variants — the base instance is the equality-constrained problem (1a)–(1d), with none of them. Figures 1 and 2 both show the $T = 6$ case. The load factor of Fig. 1 is p_L^t against a rated load of 2.0 p.u., and its price axis takes $P_{\text{BASE}} = 1$ kW, so that 1 p.u.h = 1 kWh.

grounds that with $\eta_c \eta_d < 1$ and a positive throughput cost, charging and discharging at once is never optimal. Imposing it explicitly would make it a complementarity constraint, whose Jacobian loses rank exactly at the solutions of interest — precisely the degeneracy the backward pass cannot absorb. Conversion losses are therefore cheap to add to this model only for as long as the implicit argument holds.

D. Problem Characteristics

The base instance is a strictly convex equality-constrained quadratic program: $3T$ variables, T balance equalities (1b) and $T + 1$ storage equalities (1c)–(1d). Eliminating $P_{\text{Subs}}^t = p_L^t - P_B^t$ and $B^{T+1} - B^1 = -\Delta t \sum_t P_B^t$ reduces it to the

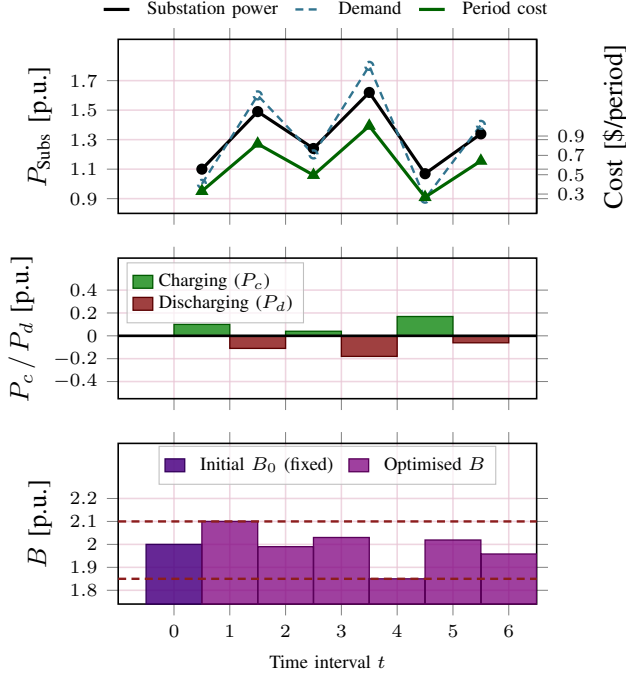


Fig. 2. The benchmark solved at $T = 6$ with all bounds (1e)–(1g) imposed. Battery actions use the convention of the tADMM study: charging is drawn upward in green and discharging downward in maroon, so a bar’s direction reads as the device’s action rather than as the sign of P_B^t . Read it as one worked example. The battery charges in the cheap intervals $t = 1, 3, 5$ and discharges into the expensive ones $t = 2, 4, 6$, which is why the substation trace in the top panel is visibly flatter than the demand it serves — it is pulled down at the two demand peaks and pushed up in the trough at $t = 5$, where energy is cheapest. Substation import is unlimited above, so the schedule is bounded only by the device: stored energy is pushed onto $B^{\max} = 2.10$ after the first charge and drawn right down onto $B^{\min} = 1.85$ after the interval-4 discharge (dashed rules in the lower panel). Those two limits are exactly the active set the independent reference solver reports, and the trajectory sits on them to 1.2×10^{-9} . The battery power bounds, $-0.5 \leq P_B^t \leq 0.30$, never bind.

unconstrained quadratic

$$J(P_B) = \text{const} - \Delta t \sum_t c^t P_B^t + C_B \Delta t \sum_t (P_B^t)^2 + w(\Delta t)^2 s^2, \quad (10)$$

with $s = \sum_t P_B^t$. Dividing stationarity through by Δt gives $-c^k + 2C_B P_B^k + 2w\Delta t s = 0$, and summing over k collapses the system to a single scalar:

$$s = \frac{\sum_t c^t}{2C_B + 2Tw\Delta t}, \quad P_B^k = \frac{c^k - 2w\Delta t s}{2C_B}. \quad (11)$$

The instance therefore admits a closed-form reference solution against which any solver applied to it can be checked exactly, which is the reason for its construction.

Storage (1d) is the only coupling between periods; without the terminal term in (1a) the objective would not depend on P_B at all, and the problem would separate into T independent single-period problems.

Reference implementation. Problem (1) is implemented in `copper_plate_battery.jl` (base instance) and `copper_plate_battery_bounds.jl` (with (1e)–(1g)), both under `ddp/examples/power_system/`. Both carry Δt symbolically, so the implementation matches (1) term

for term. One choice is worth flagging: C_B is set to 0.5 here, whereas the tADMM paper uses $C_B \approx 10^{-6} \min_t c^t$, deliberately near-zero so that battery throughput cannot compete with energy cost. Since C_B is exactly the reduced-Hessian curvature, that near-zero value makes the instance badly conditioned by construction; a healthy C_B is used for the base test, and shrinking it toward the paper’s value is better treated as a deliberate conditioning study than as the default.

REFERENCES

- [1] A. R. Jha, S. Paul, and A. Dubey, “Analyzing the performance of linear and nonlinear multi-period optimal power flow models for active distribution networks,” in *2025 IEEE North-East India International Energy Conversion Conference and Exhibition (NE-IECCCE)*. IEEE, 2025, pp. 04–06.